THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 8 — MATHS · MEDIUM LEVEL CHAPTER TEST SET 3

Chapter 13: **Algebra Play** | RBSE Pattern | Session 2026–27

Time: 1 Hour 30 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 40**

General Instructions:

1. All questions are compulsory.
2. Q. 1–10 are Objective Type (1 mark each).
3. Q. 11–14 are Very Short Answer Type (2 marks each).
4. Q. 15–18 are Short Answer Type (3 marks each).
5. Q. 19–20 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 17 to 20.
7. Write the question number before each answer.
8. Medium level: multi-step questions.

SECTION — A (Objective Type Questions) 10 × 1 = 10

Choose the correct option and write it in your answer book.

1. If $a = 15$ and $b = 2$, the value of $2a - 9b + 5$ is: 1

(a) 15 **(b)** 34 **(c)** 17 **(d)** 19

2. A rectangle has length $(2x + 7)$ cm and breadth $(6x + 2)$ cm. Its perimeter (cm) is: 1

(a) $16x + 18$ **(b)** $12x^2 + 14$ **(c)** $16x + 9$ **(d)** $8x + 9$

3. If $x + y = 7$ and $xy = 12$, then $x^2 + y^2$ equals: 1

(a) 49 **(b)** 25 **(c)** 73 **(d)** 37

4. The sum of 6 consecutive even numbers starting with $2n$ is: 1

(a) $6n + 30$ **(b)** $12n + 32$ **(c)** $12n + 30$ **(d)** $12n + 6$

5. On expanding $7(x + 2) + 9x$ we get: 1

(a) $16x + 2$ **(b)** $63x + 14$ **(c)** $7x + 23$ **(d)** $16x + 14$

6. $(x + 5)(x + 8)$ equals: 1

(a) $x^2 + 13x + 40$ **(b)** $2x + 13$ **(c)** $x^2 + 40x + 13$ **(d)** $x^2 + 13$

7. If $a = 10$ and $b = 7$, the value of $5a - 4b + 12$ is: 1

(a) 31 **(b)** 34 **(c)** 68 **(d)** 37

8. A rectangle has length $(6x + 5)$ cm and breadth $(2x + 3)$ cm. Its perimeter (cm) is: 1

(a) $16x + 8$ **(b)** $16x + 16$ **(c)** $12x^2 + 15$ **(d)** $8x + 8$

9. If $x + y = 15$ and $xy = 44$, then $x^2 + y^2$ equals: 1

(a) 225 **(b)** 137 **(c)** 181 **(d)** 313

10. The sum of 7 consecutive even numbers starting with $2n$ is: 1

(a) $14n + 7$ **(b)** $7n + 42$ **(c)** $14n + 44$ **(d)** $14n + 42$

SECTION — B (Very Short Answer Type) 4 × 2 = 8

Answer in about 20–30 words each.

11. A two-digit number has tens digit a and ones digit b, and $a - b = 4$. Show that the number minus its reverse is 36. 2

12. Expand and simplify $3(x + 7) + 6(x - 2)$. 2

13. Expand $(x + 2)(x + 11)$. 2

14. A two-digit number has tens digit a and ones digit b, and $a - b = 5$. Show that the number minus its reverse is 45. 2

SECTION — C (Short Answer Type) 4 × 3 = 12

Answer in about 40–60 words each.

15. The sides of a triangle are $(5x + 2)$ cm, $(2x + 7)$ cm and $(3x + 3)$ cm. Find its perimeter, and its value when $x = 6$. 3

16. If $x + y = 16$ and $xy = 55$, find $x^2 + y^2$. 3

17. Show that the sum of 7 consecutive even numbers starting with $2n$ is $14n + 42$ and verify it for $n = 5$. 3

OR

The sides of a triangle are $(2x + 6)$ cm, $(4x + 3)$ cm and $(5x + 1)$ cm. Find its perimeter, and its value when $x = 7$. 3

18. If $x + y = 17$ and $xy = 60$, find $x^2 + y^2$. 3

OR

Show that the sum of 3 consecutive even numbers starting with $2n$ is $6n + 6$ and verify it for $n = 9$. 3

SECTION — D (Long Answer Type) 2 × 5 = 10

Answer in about 100–150 words each.

19. Expand and simplify $(x + 5)(x + 5) - x(x + 2)$. Find its value at $x = 6$. 5

OR

Think of a two-digit number with digits a and b. Add its reverse. Show that the sum is 11(a+b) and find the sum for the digits 7, 9. Is the sum divisible by 11? 5

20. Expand and simplify $(x + 9)(x + 3) - x(x + 7)$. Find its value at $x = 2$. 5

OR

Think of a two-digit number with digits a and b. Add its reverse. Show that the sum is 11(a+b) and find the sum for the digits 1, 9. Is the sum divisible by 11? 5

END OF QUESTION PAPER Total: 40 Marks

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